Quantum Random Walks

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1 Quantum Walk Framework

Let's delve into the mathematical details of the quantum walk framework described in the excerpt.

1.1 State Representation

The state of the quantum walk is represented as a superposition of position states (nodes) and coin states. Mathematically, it can be represented as:

$$|\psi_i\rangle = \sum_{j=1}^{N} \sum_{k=1}^{N} A_{j,k} |j,k\rangle$$

Where:

- $|\psi_i\rangle$ is the state of the walker at time i.
- $|j,k\rangle$ represents the state where the walker is at node j with coin state k.
- $A_{j,k}$ are complex amplitudes.

1.2 Coin Operator

The coin operator, denoted by $\hat{C}_i$, is a unitary transformation that acts on the coin states at each node. It represents the coin flip operation. In the case where the coin states are represented by a $N \times N$ matrix, $\hat{C}_i$ is an $N \times N$ unitary matrix.

1.3 Translation Operator

The translation operator, denoted by $\hat{T}_i$, describes how the walker moves from one node to adjacent nodes based on the outcome of the coin flip. It is a conditional translation operator that moves the walker according to some predefined rule.

1.4 Quantum Walk Evolution

The quantum walk evolves through repeated applications of the coin operator followed by the translation operator. Mathematically, the evolution of the quantum walk after n steps can be represented as:

$$|\psi_n\rangle = \hat{T}_n \hat{C}_n \ldots \hat{T}_2 \hat{C}_2 \hat{T}_1 \hat{C}_1 |\psi_0\rangle$$

Where:

- $|\psi_n\rangle$ is the state of the walk after n steps.
- $\hat{C}_i$ and $\hat{T}_i$ are the coin and translation operators at the i-th step.
- $|\psi_0\rangle$ is the initial state of the walker.

1.5 Grid Representation

The Hilbert space of the quantum walk can be represented as a two-dimensional grid, where each row/column corresponds to a node of the graph, and each grid element represents a combination of a node and a coin state.

1.6 Modification for General Graphs

To generalize the quantum walk to arbitrary graphs, unwanted edges are removed from the complete graph. Instead of removing corresponding states from the Hilbert space, the coin operators are modified to isolate unwanted states from interacting with other states.

The modification ensures that unwanted states, corresponding to removed edges, do not interact with other states during the quantum walk.

Overall, the mathematical framework provides a rigorous foundation for studying quantum dynamics on graphs and can be applied to analyze various quantum algorithms and processes.

2 Physical Implementation using Quantum Dots

Let's delve into the mathematical aspects of the proposed physical implementation of the quantum walk using a two-dimensional grid of quantum dots.

2.1 State Representation

The state of the quantum walk is represented by a wavefunction, typically denoted by $|\psi\rangle$, which describes the probability amplitude of finding the electron at each quantum dot in the grid.

2.2 Grid Construction

The grid is constructed as a $2N \times 2N$ grid, where N represents the number of nodes in the quantum walk. Each quantum dot in the grid corresponds to a specific node in the quantum walk.

2.3 Quantum Dot Interactions

The evolution of the quantum walk involves manipulating the electron wavefunction throughout the grid by introducing appropriate quantum dot interactions.

2.4 CS Decomposition

The arbitrary N-level unitary operators $\hat{c}_H^j$ and $\hat{c}_V^j$ (representing coin operations along rows and columns) can be implemented using a CS (Cayley-Sudakov) decomposition. This decomposition breaks down a general N-level rotation matrix into a series of pair-wise or qubit rotations.

2.5 Pair-Wise Interactions

The pair-wise interactions between neighboring quantum dots correspond to the action of the unitary operators along rows and columns. These interactions can be represented mathematically using matrices that describe the transformations induced by the interactions.

2.6 Non-Neighboring Interactions

To extend the implementation to non-neighboring quantum dots, a systematic form of interaction is introduced. This involves defining a systematic pattern of interactions between dots that are not immediate neighbors but are required for implementing the desired rotations.

2.7 Mathematical Formulation

1) **Pair-Wise Interaction**: The pair-wise interaction between neighboring quantum dots can be represented mathematically using unitary matrices that describe the transformations induced by the interactions. The unitary matrices capture the evolution of the electron wavefunction as it moves between neighboring dots.
2) **Extension to Non-Neighboring Dots**: Extending the interaction scheme to non-neighboring dots involves defining a mathematical framework that accounts for the systematic pattern of interactions. This may involve defining transformation matrices that describe the interactions between non-neighboring dots based on the predefined pattern.

2.8 Optimization and Validation

The mathematical formulation needs to be optimized and validated to ensure that it accurately captures the dynamics of the quantum walk. Optimization may involve minimizing computational complexity or resource usage, while validation involves testing the mathematical framework through simulations or experimental implementations.
In summary, the mathematical aspects of the proposed physical implementation involve representing the quantum walk dynamics using unitary transformations and defining interaction patterns that facilitate the evolution of the electron wavefunction across the grid of quantum dots.

3 Conclusion

The article discusses the implementation of quantum random walks using electrons trapped in a 2D array of quantum dots. The process involves manipulating the electron wavefunction across the grid of quantum dots to simulate the dynamics of the quantum random walk.

3.1 Pair-wise Interactions

1) **Rotations Between Neighboring Dots**:
   - A pair of neighboring quantum dots can undergo a unitary rotation by controlling the potential barrier between them.
   - When the barrier is lowered, the electron wave packet can move between the two dots.
   - By precisely timing the raising of the barrier, the electron can be recaptured with a desired distribution across the two dots.
2) **Rotations Between Non-Neighboring Dots**:
   - To interact a pair of non-neighboring quantum dots, the dots must be moved close to each other, the desired rotation applied, and then returned to their original locations.
   - This process is systematic and viable due to the specific pattern of pairwise interactions defined by Eq. 3.

3.2 Quantum Dot Conveyor Belt

1) Apply $N/2$ simultaneous π rotations to pairs of quantum dots and their adjacent register dots to transfer electron wave packets to the register dots.
2) Adiabatically move the register quantum dots, effectively carrying the electron wave packets along the register row/column.
3) Apply $N/2$ general rotations to the newly coupled quantum dot pairs.
4) Return the register dots to their original location by reversing the adiabatic motion.
5) Introduce another π rotation to move the amplitudes from register dots back to their original positions.

3.3 Modeling Qubit Rotations

- The time-dependent Hamiltonian governing the quantum dynamical evolution of the system is solved using the Schrödinger equation.
- The equation is solved efficiently using a Chebychev polynomial expansion.
- By controlling the central potential barrier and/or electron-electron interactions, all possible qubit rotations in the Bloch sphere can be achieved.

3.4 Conclusion

In summary, the mathematical framework involves solving the time-dependent Schrödinger equation using a Chebychev polynomial expansion to simulate the dynamics of electrons trapped in a 2D array of quantum dots. Qubit rotations are achieved by precisely controlling the time-dependent Hamiltonian, leading to arbitrary quantum gate operations.